Worksheet-3 (Solutions)

(1). Any CDF F has the following properties:

(i) F is non-decreasing, i.e., $x_1 \leq x_2 \Rightarrow F(x_1) \leq F(x_2)$

(ii) F is right continuous, i.e., $F(a) = \lim_{x \to a^+} f(x)$

(iii) $\lim_{x \to \infty} F(x) = 1$ & $\lim_{x \to -\infty} F(x) = 0$.

(a) $$F_1(c) = \begin{cases} 0 & , c \leq 0 \\ 0.5 & , 0 < c \leq 1 \\ 0.75 & , 1 < c \leq 3 \\ 1 & , c > 3 \end{cases}$$

- $F_1(c)$ is non decreasing ✓
- $\lim_{c \to -\infty} F_1(c) = 0$ & $\lim_{c \to \infty} F_1(c) = 1$. ✓

$F_1(0) = 0$, $\lim_{c \to 0^+} F_1(c) = 0.5$

$\Rightarrow$ ~~F~~ $F_1(c)$ is not right continuous at $c = 0$

$\Rightarrow$ Not a CDF.

(b) $$F_2(c) = \begin{cases} 0 & , c \leq 0 \\ 0.5c & , 0 < c \leq 1 \\ 0.25 + 0.25c & , c > 1 \end{cases}$$

Non-decreasing

let $c_1 \in (0,1]$ & $c_2 \in (1, \infty)$

$\Rightarrow F_2(c_1) = 0.5 c_1 = \frac{c_1}{2}$, $F_2(c_2) = \frac{1}{4} + \frac{c_2}{4} = \frac{c_2+1}{4}$

To show: $F_2(c_1) \leq F_2(c_2)$

$\Leftrightarrow \frac{c_1}{2} \leq \frac{c_2+1}{4} \Leftrightarrow 2c_1 \leq c_2 + 1$

which is true as $2c_1 \in (0,2]$ & $c_2 + 1 \in (2, \infty)$.

Rest cases are easy to check!

Right continuous

$$\lim_{c \to 0^+} F_2(c) = 0.5 \times 0 = 0 = F_2(0).$$

$$\lim_{c \to 1^+} F_2(c) = 0.25 + 0.25 \times 1 = 0.5 = 0.5 \times 1 = F_2(1).$$

$\Rightarrow F_2(c)$ is right continuous on $\mathbb{R}$.

$\lim_{c \to -\infty} F_2(c) = 0$, but $\lim_{c \to \infty} F_2(c) = \lim_{c \to \infty} (0.25 + 0.25c) = \infty$

$\Rightarrow F_2(c)$ is not a CDF.

(c)

$$F_3(c) = \begin{cases} 0, & c < 0 \\ 0.5, & 0 \le c < 1 \\ 0.25, & 1 \le c < 3 \\ 1, & c > 3. \end{cases}$$

$0.5 < 2$ but $F_3(0.5) = 0.5 \not\leq F_3(2) = 0.25$

$\Rightarrow$ F_3 is <u>not</u> non-decreasing, i.e., $a \leq b \not\Rightarrow F_3(a) \leq F_3(b)$.

$\Rightarrow$ $F_3(c)$ is not a CDF.

(2)

Probab of success $= p$.
probab of feilure $= 1-p$.

(a) X_n : # of successes achieved in the first n clicks, i.e., during the time interval $t \in \{1, 2, \cdots, n\}$

Probability that there are k success in first n clicks (independent trials → Binomial dist)

$$p_{X_n}(k) = P(X_n = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, \cdots, n.$$

(b) $P(X_n = k \mid X_m = i)$, $n > m$

probab that there are k successes in first n clicks, given that there are i successes in the first m clicks.

=~~probab that~~.

Given $X_m = i$, the first ~~n~~ m trials have i successes. For $X_n = k$, the # of successes in the first n trials is k, which means: In trials $m+1, m+2, \cdots, n$, we need $k-i$ successes.

$\because$ trials are independent, the # of successes in these $n-m$ trials is independent of the earlier m trials, which follows binomial distribution with $n-m$ trials & probab of success is p.

$$\therefore P(X_n = k \mid X_m = i) = \binom{n-m}{k-i} p^{k-i} (1-p)^{(n-m)-(k-i)}, \quad k = i, i+1, \cdots, (n-m)+i$$

(c) $P(X_m = i \mid X_n = k)$

Probab that there are i successes in the first m clicks, given that there are k successes in first n clicks, where $m<n$

By defn of conditional probab.

$$P(X_m = i \mid X_n = k) = \frac{P(X_m = i,\ X_n = k)}{P(X_n = k)}$$

but,

$$P(X_m = i,\ X_n = k) = P(i \text{ successes in } m \text{ trials}) \cdot P(k-i \text{ successes in } n-m \text{ trials})$$

$$= \binom{m}{i} p^i (1-p)^{m-i} \cdot \binom{n-m}{k-i} p^{k-i} (1-p)^{n-m-(k-i)}$$

$$P(X_n = k) = \binom{n}{k} p^k (1-p)^{n-k}.$$

$$\Rightarrow P(X_m = i \mid X_n = k) = \frac{\binom{m}{i} p^k (1-p)^{n-k} \binom{n-m}{k-i}}{\binom{n}{k} p^k (1-p)^{n-k}} = \frac{\binom{m}{i}\binom{n-m}{k-i}}{\binom{n}{k}},$$

$$\max\{0, k-(n-m)\} \le i \le \min\{m, k\}$$

(d) T_1: Time at which the 1st success occurs.

$P(T_1 = u)$ = Probability that the first success happens on the u^{th} click.

$$= P(\text{failure})^{u-1} \cdot P(\text{success})$$

$$= (1-p)^{u-1} \cdot p = p(1-p)^{u-1} \quad =.$$

(3) For a valid probability distribution

$$\sum_x P(x) = 1 \quad \& \quad P(x) \geq 0 \ \forall x$$

(a)

x	5	10	15
$P(x)$	0.3	0.4	0.1

$$\sum_x P(x) = P(5) + P(10) + P(15)$$
$$= 0.3 + 0.4 + 0.1$$
$$= 0.8 \neq 1$$

$\Rightarrow$ Not a valid probab. dist.

(b)

x	8	12	16	20
$P(x)$	$\frac{5}{12}$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$

$$\sum_x P(x) = P(8) + P(12) + P(16) + P(20)$$
$$= \frac{5}{12} + \frac{1}{12} + \frac{1}{12} + \frac{1}{12}$$
$$= \frac{8}{12} = \frac{2}{3} \neq 1$$

$\Rightarrow$ Not a valid probab dist.